

Filters

Select Year

2020 × 2021 × 2022 ×

2023 × 2024 ×

Select Region

Choose an option



Telemedicine Operations Intelligence Platform

Cloud-based healthcare analytics platform powered by AWS S3, Athena, Python, Streamlit, and FastAPI

AWS Athena access failed. Loading local CSV fallback for demo.

Total Enrollment

595,276,985

Total Telehealth Users

188,949,348

Avg Telehealth Rate

33.31%

📌 **Platform Purpose:** This dashboard helps healthcare leaders monitor telehealth adoption, identify regional disparities, and support data-driven planning for digital healthcare access.



Strategic Insights for Healthcare Decision-Makers

- In 2024, 4 regions show high telehealth adoption above 50%.
- 47 regions remain below 25% adoption in the latest selected year.
- Across the selected period, 12 regions have an average adoption rate below 25%.
- These gaps may indicate differences in digital access, provider availability, broadband infrastructure, or regional healthcare readiness.



Policy and Business Recommendation

Healthcare organizations and policymakers should prioritize regions with persistently low telehealth adoption.

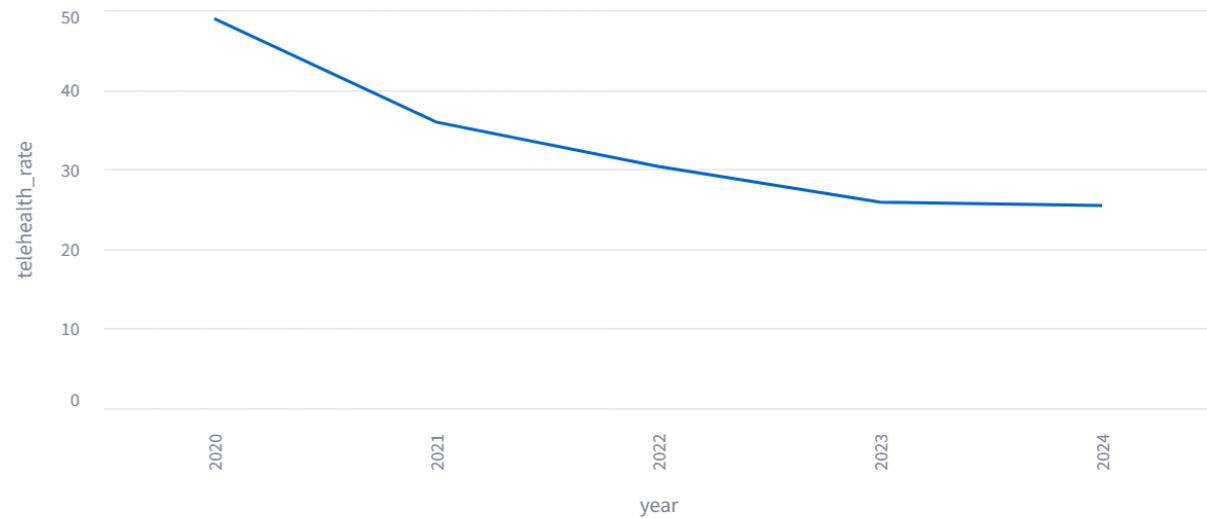
Recommended actions:

- Expand broadband and digital infrastructure
- Improve patient digital literacy
- Incentivize telehealth provider participation
- Monitor adoption trends continuously
- Use regional analytics to guide resource allocation

This supports healthcare access equity and helps institutions make evidence-based operational decisions.



Telehealth Adoption Trend



From 2020 to 2024, the average telehealth adoption rate changed from 48.91% to 25.46%, representing a change of -23.45 percentage points.



Region-Level Machine Learning Forecast

Model Performance Comparison

- Linear Regression MAE: 7.31 percentage points
- Random Forest MAE: 4.97 percentage points

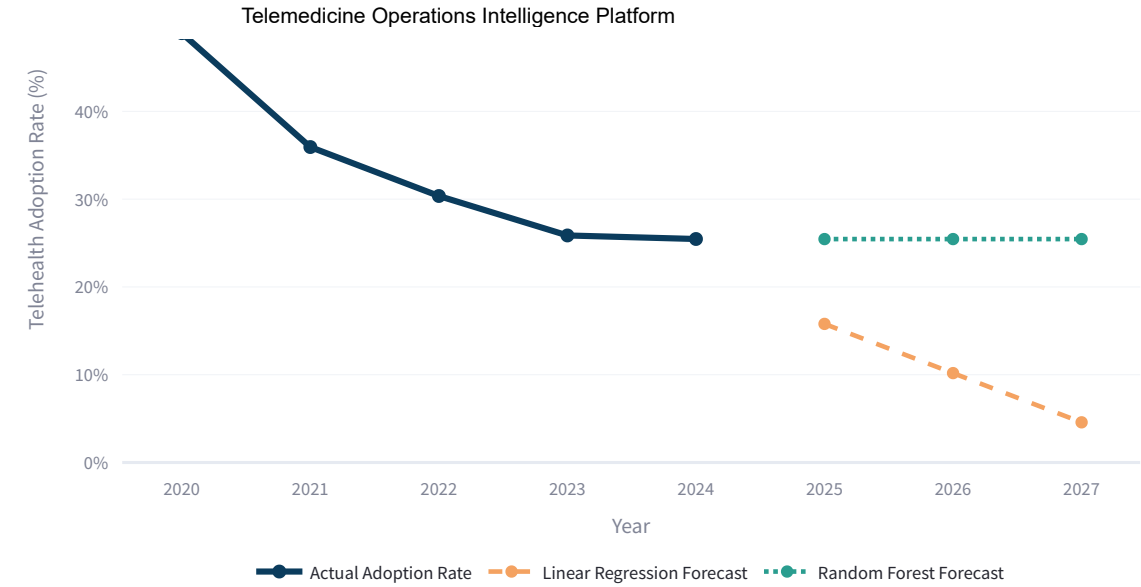
Best performing model: **Random Forest Regressor**

Telehealth Adoption Forecast: Actual vs Model-Based Projections

60%

50%





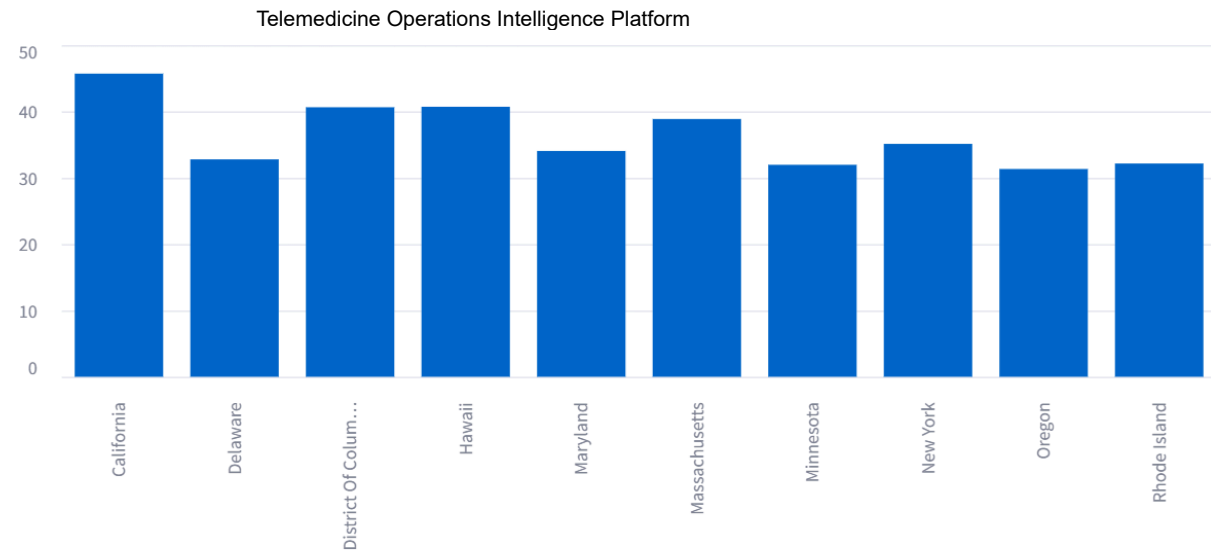
Executive Forecast Insight:

The model comparison indicates that **Random Forest Regressor** performs better based on MAE. Under current regional patterns, average telehealth adoption is projected to reach approximately **25.45%** by **2027**.

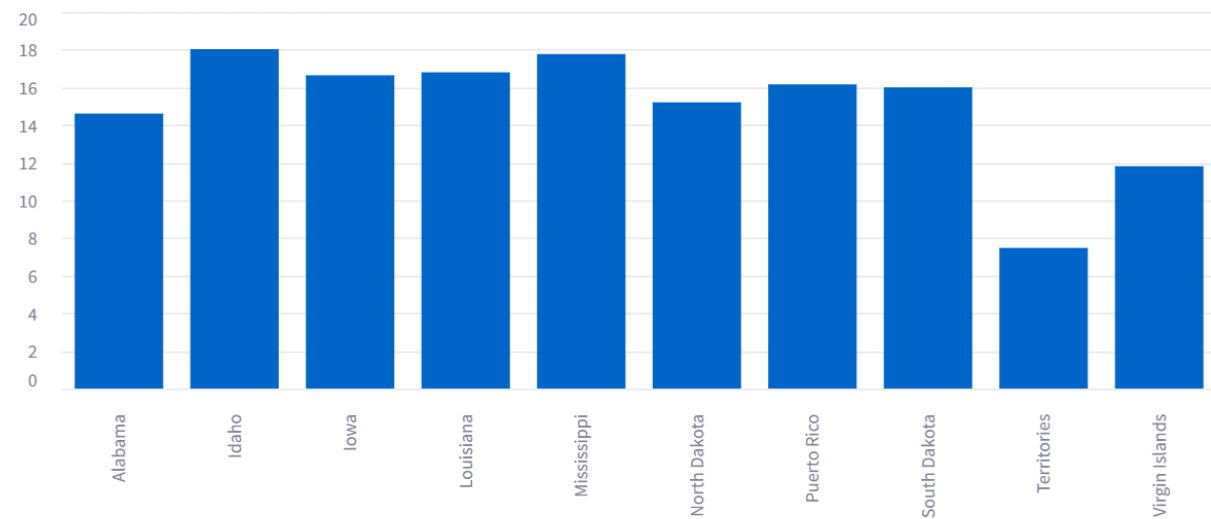
View regional forecast table

⚠️ Note: This forecast is a prototype decision-support feature. Although the dataset contains many observations, predictions are based on limited historical years and regional patterns. Future work should incorporate higher-frequency data, additional healthcare access variables, and advanced time-series models before operational use.

🏆 Top 10 Regions by Telehealth Adoption — 2024



⚠️ Lowest 10 Regions by Telehealth Adoption — 2024



Adoption Gap Analysis

Adoption Gap: Top 10 vs Lowest 10 Regions

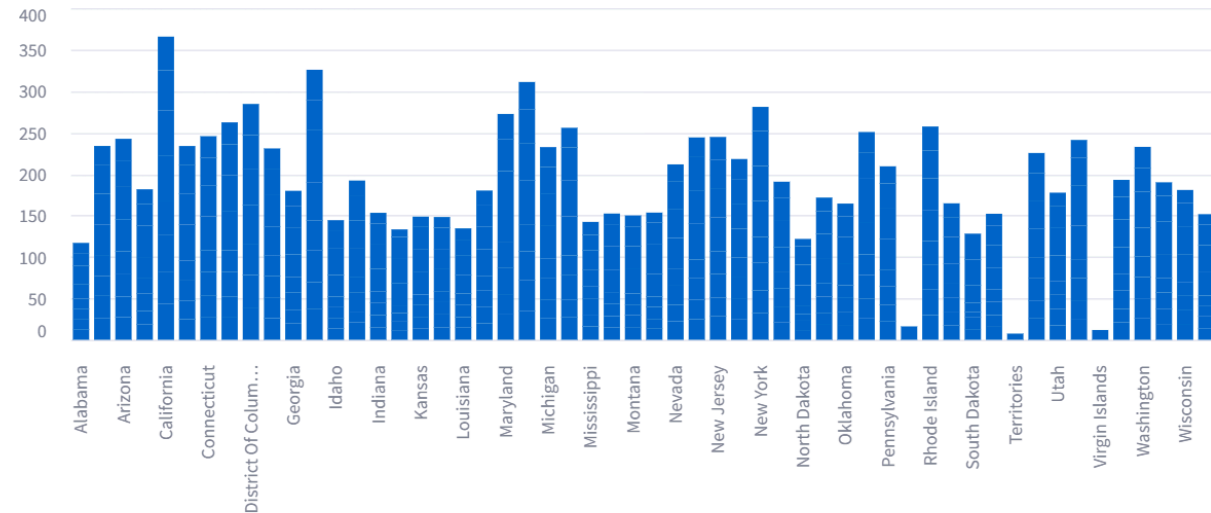
21.27 percentage points

The average adoption rate among the top 10 regions is **36.32%**, while the average among the lowest 10 regions is **15.05%**.

This gap highlights regional inequality in telehealth adoption and helps institutions identify where access improvement may be needed.



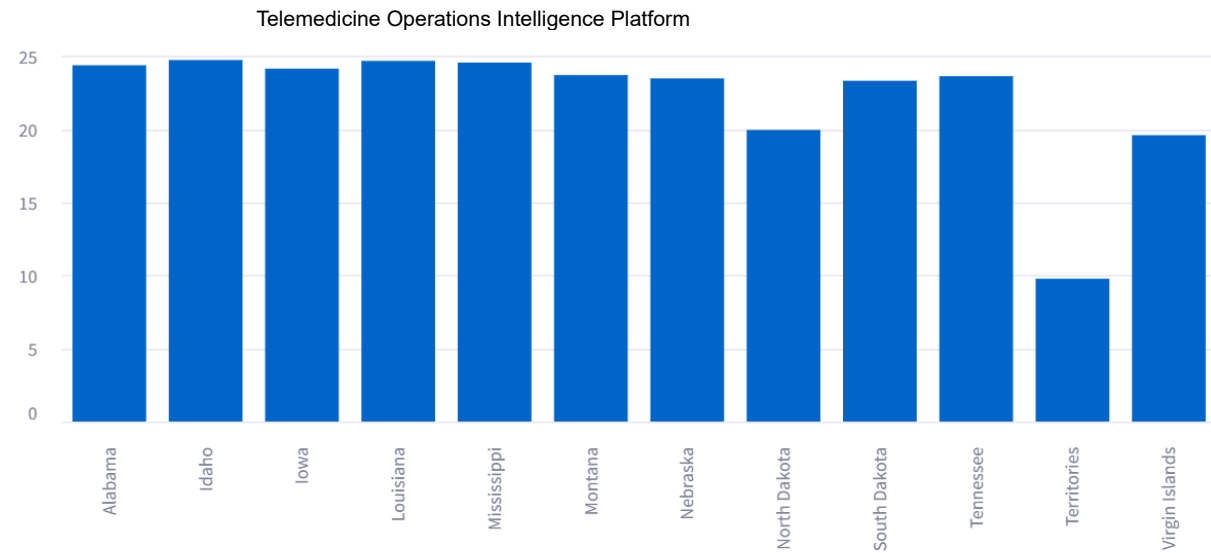
Telehealth Adoption Distribution (Equity View)



This equity view shows how telehealth adoption is distributed across regions in the latest selected year. It helps identify whether adoption is broadly shared or concentrated among a smaller group of higher-performing regions.



At-Risk Regions with Low Average Adoption



12 regions have an average telehealth adoption rate below 25% across the selected period. These regions may require targeted support to improve digital healthcare access.



Regional Telehealth Trend Comparison

Select one or more regions from the sidebar to compare regional trends.



Data Preview

	year	quarter	region	total_enrollment	telehealth_users	telehealth_rate
4	2,020	Overall	California	2,937,567.8333	1,602,926	59.69
5	2,020	Overall	Colorado	448,799.9167	191,989	45.72
6	2,020	Overall	Connecticut	315,203.25	181,962	58.61
7	2,020	Overall	Delaware	159,145.1667	96,701	61.62
8	2,020	Overall	District Of Columbia	57,945.9167	31,743	58.81
9	2,020	Overall	Florida	2,120,377.4167	1,000,400	48.17
10	2,020	Overall	Georgia	882,774.9167	350,406	40.87
11	2,020	Overall	Hawaii	110,054.1667	52,125	51.31
12	2,020	Overall	Idaho	194,966.3333	61,565	33.68
13	2,020	Overall	Illinois	1,386,115.0833	610,824	45.74

 Live Model Prediction via API

Generate real-time telehealth forecasts using the deployed FastAPI model.

Forecast Year

2027

-

+

Total Enrollment

1000000.00

-

+

Region

Alabama

▼

Telehealth Users

250000.00

-

+

 Generate API Forecast

Prediction generated successfully

Forecasted Telehealth Rate

23.24%

Calculated Current Rate

25.00%

```
▼ {  
  "year" : 2027
```

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"region" : "Alabama"  
"total_enrollment" : 1000000   
"telehealth_users" : 250000  
"calculated_input_rate" : 25  
"forecasted_telehealth_rate" : 23.24  
}
```